PHYSICS I
WEEK 12

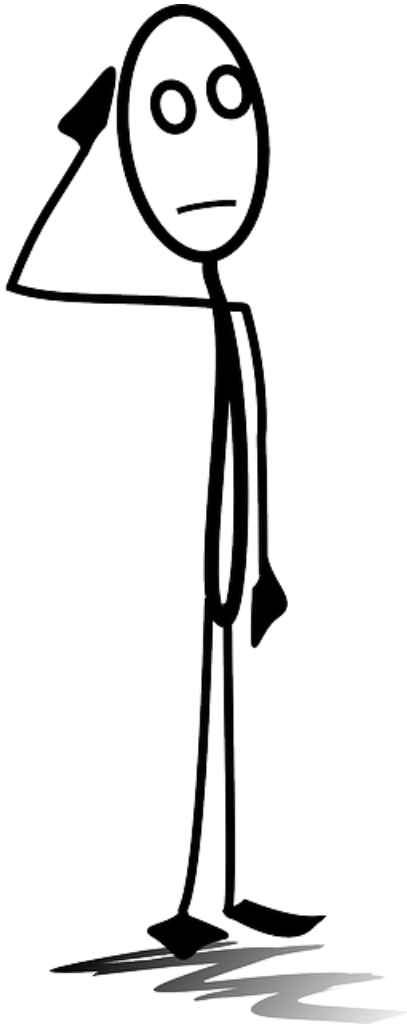


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WEEK 12: OUTLINE

- Motion under Central Forces:
  - Lagrangian in plane polar coordinates:

    Morin 7.1, 7.2
  - Gravity: Morin 7.3, 7.4

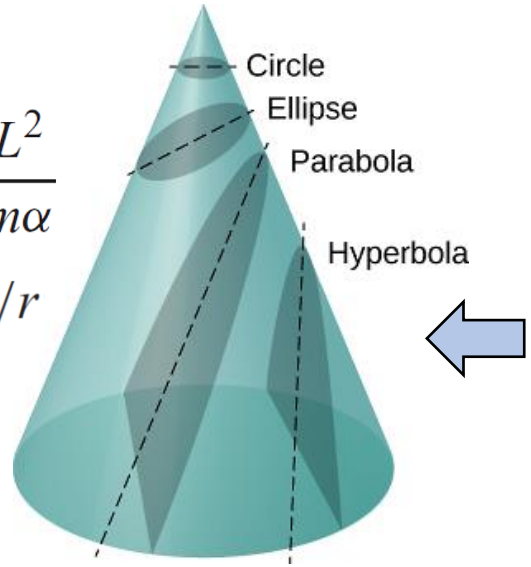


CENTRAL FORCE: GRAVITY

Finding $r(\theta)$

$$\left(\frac{1}{r^2}\frac{dr}{d\theta}\right)^2 = \frac{2mE}{L^2} - \frac{1}{r^2} - \frac{2mV(r)}{L^2}$$

$$V(r) = -\frac{\alpha}{r}, \quad \text{where } \alpha \equiv GM_{\odot}m$$



$$k \equiv \frac{L^2}{m\alpha}$$

$$\cos\theta = x/r$$

$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon\cos\theta),$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}}$$

$$r_{\min} = \frac{L^2}{m\alpha(1+\epsilon)}$$

$$r_{\max} = \frac{L^2}{m\alpha(1-\epsilon)} \quad (\text{if } \epsilon < 1)$$

$$r_{\max} = \infty \quad (\text{if } \epsilon \geq 1)$$

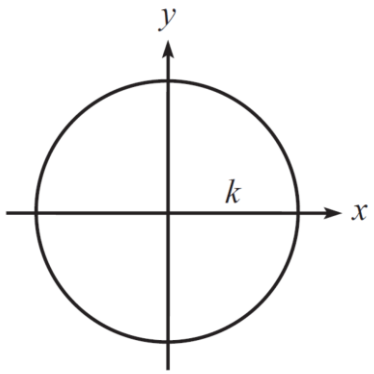
$$x^2 + y^2 = k^2 - 2k\epsilon x + \epsilon^2 x^2$$

CENTRAL FORCE: GRAVITY

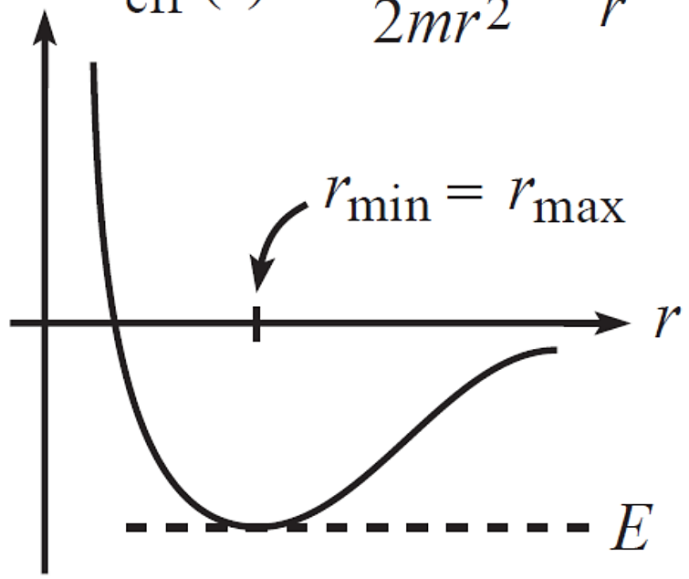
- **Circle** ($\epsilon = 0$)

$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon \cos\theta), \quad \Rightarrow \quad x^2 + y^2 = k^2$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}} \quad \Rightarrow \quad E = -m\alpha^2/2L^2$$



$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{\alpha}{r}$$



$$r_{\min} = r_{\max} = L^2/m\alpha$$

CENTRAL FORCE: GRAVITY

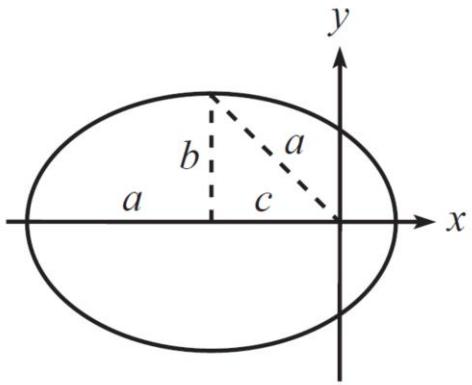
- **Ellipse** $(0 < \epsilon < 1)$

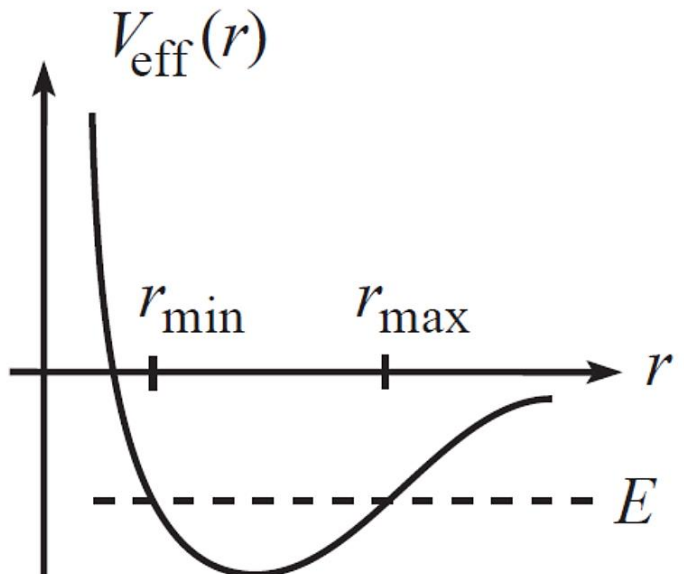
$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon\cos\theta),$$

$$\frac{\left(x + \frac{k\epsilon}{1-\epsilon^2}\right)^2}{a^2} + \frac{y^2}{b^2} = 1,$$

$$a = \frac{k}{1-\epsilon^2}, \quad \text{and} \quad b = \frac{k}{\sqrt{1-\epsilon^2}}$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}} \quad \Rightarrow \quad -m\alpha^2/2L^2 < E < 0$$





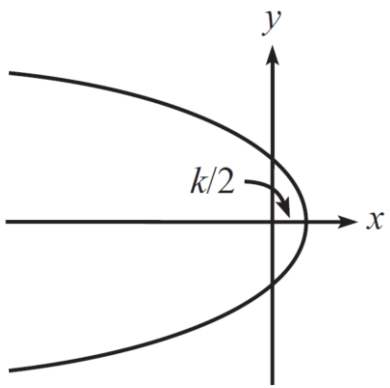
$$r_{\text{min}} = \frac{L^2}{m\alpha(1+\epsilon)} \qquad r_{\text{max}} = \frac{L^2}{m\alpha(1-\epsilon)}$$

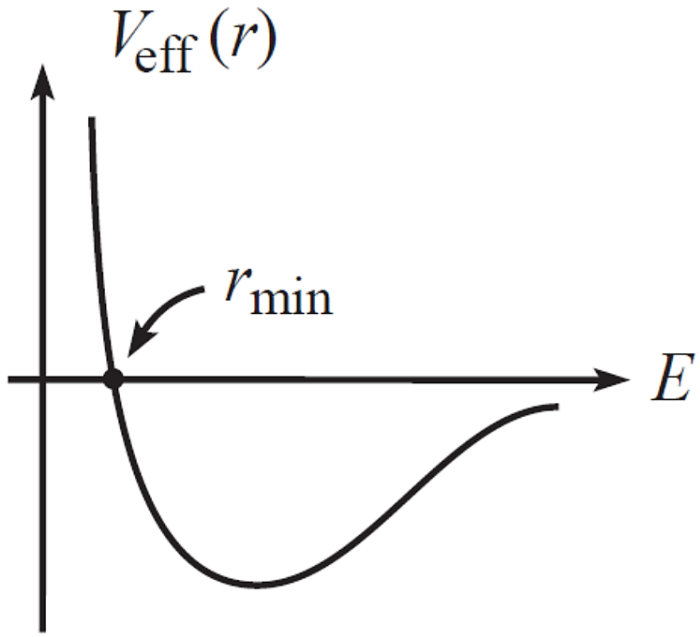
CENTRAL FORCE: GRAVITY

- **Parabola** ($\epsilon = 1$)

$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon\cos\theta), \quad \Rightarrow \quad y^2 = k^2 - 2kx$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}} \quad \Rightarrow \quad E = 0$$





$$r_{\text{min}} = L^2/2m\alpha \qquad r_{\text{max}} = \infty$$

This value of E implies that the particle barely makes it out to infinity (its speed approaches zero as $r \rightarrow \infty$)

CENTRAL FORCE: GRAVITY

- **Hyperbola** ($\epsilon > 1$)

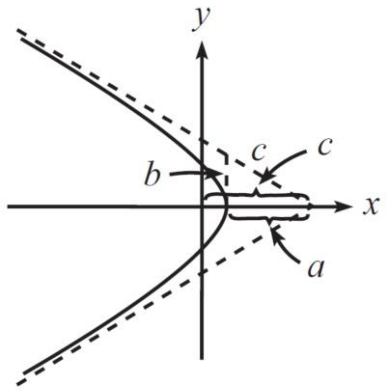
$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon\cos\theta),$$

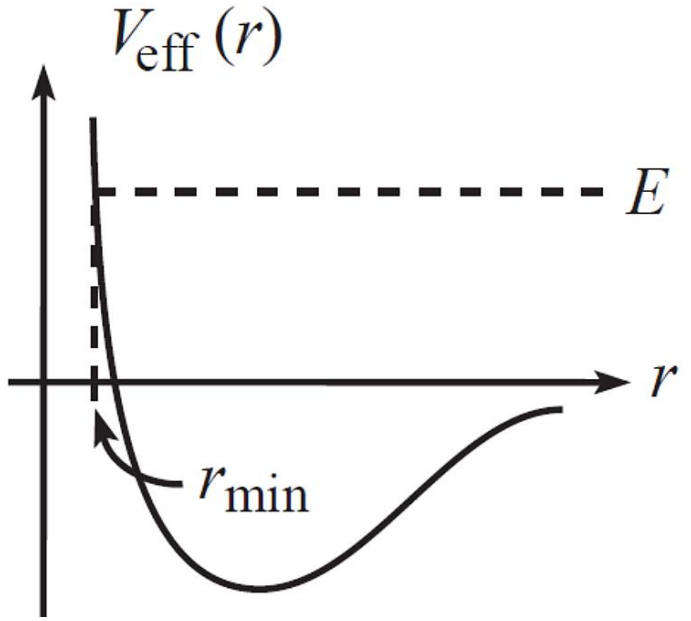
$$\frac{\left(x - \frac{k\epsilon}{\epsilon^2-1}\right)^2}{a^2} - \frac{y^2}{b^2} = 1,$$

$$a = \frac{k}{\epsilon^2 - 1}, \quad \text{and} \quad b = \frac{k}{\sqrt{\epsilon^2 - 1}}$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}}$$

$$E > 0$$





$$r_{\text{min}} = \frac{L^2}{m\alpha(1+\epsilon)} \qquad r_{\text{max}} = \infty$$

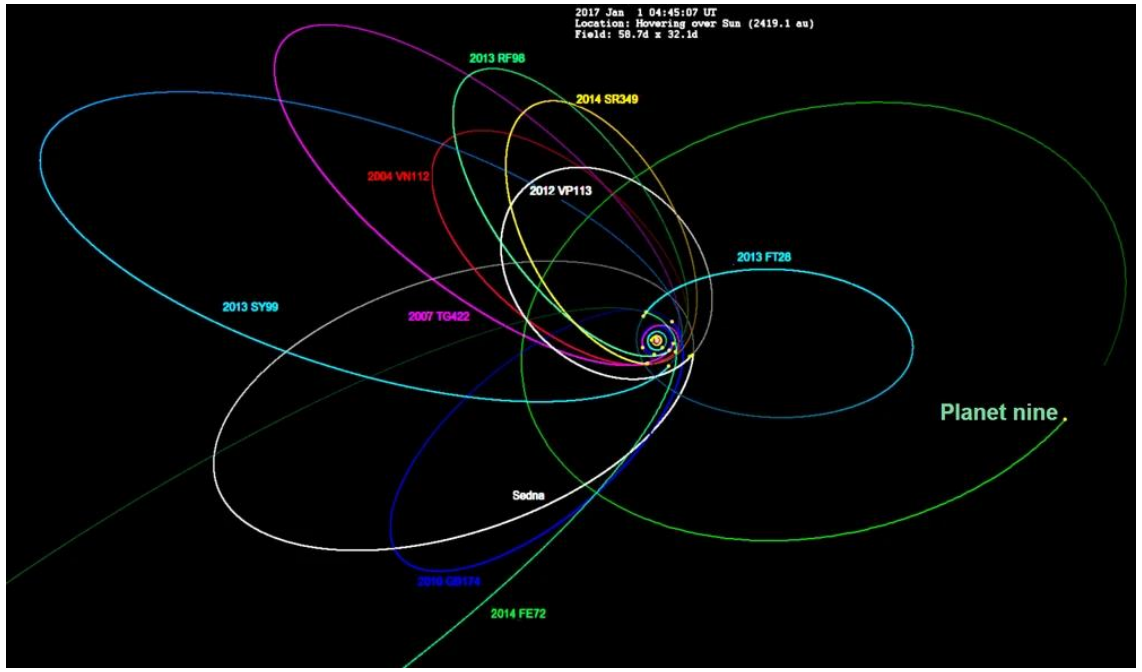
CENTRAL FORCE: KEPLER'S LAWS

- **First law:** *The planets move in elliptical orbits with the sun at one focus.*

$$\frac{\left(x + \frac{k\epsilon}{1-\epsilon^2}\right)^2}{a^2} + \frac{y^2}{b^2} = 1,$$

$$a = \frac{k}{1-\epsilon^2}, \quad \text{and} \quad b = \frac{k}{\sqrt{1-\epsilon^2}}$$

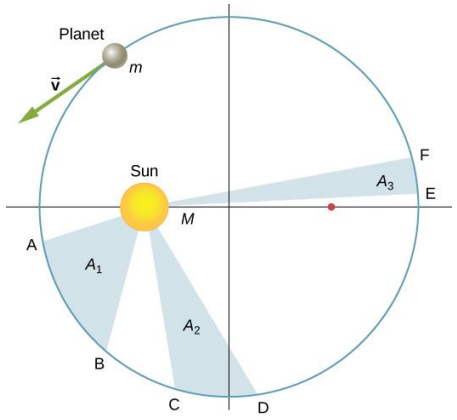
$$k \equiv \frac{L^2}{m\alpha} \qquad \alpha \equiv GM_{\odot}m$$

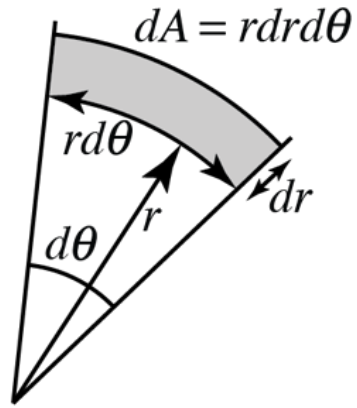


CENTRAL FORCE: KEPLER'S LAWS

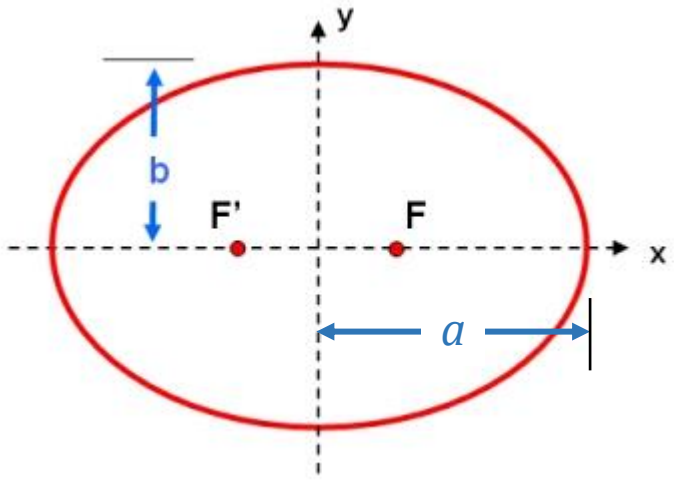
- **First law:** *The planets move in elliptical orbits with the sun at one focus.*

- **Second law:** *The radius vector to a planet sweeps out area at a rate that is independent of its position in the orbit.*





CENTRAL FORCE: KEPLER'S LAWS



$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad \Rightarrow \quad r=\frac{a(1-\epsilon^2)}{1+\epsilon\, cos\theta}$$

$$Area=\pi ab$$

$$\epsilon=\sqrt{1-\frac{b^2}{a^2}}$$

$$\frac{1}{r}=\frac{m\alpha}{L^2}(1+\epsilon\cos\theta),$$

CENTRAL FORCE: KEPLER'S LAWS

- **First law:** *The planets move in elliptical orbits with the sun at one focus.*

- **Second law:** *The radius vector to a planet sweeps out area at a rate that is independent of its position in the orbit.* $$\frac{dA}{dt} = \frac{r^2\dot{\theta}}{2} = \frac{L}{2m}$$

- **Third law:** *The square of the period of an orbit, T, is proportional to the cube of the semi-major-axis length, a.*

CENTRAL FORCE: KEPLER'S LAWS

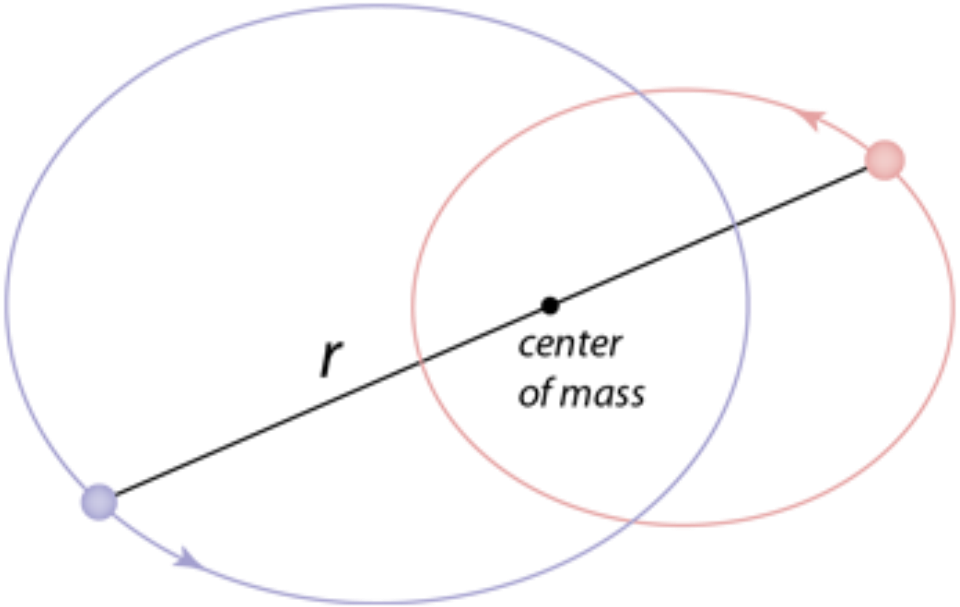
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- **Third law:** *The square of the period of an orbit, T, is proportional to the cube of the semi-major-axis length, a. More precisely,*

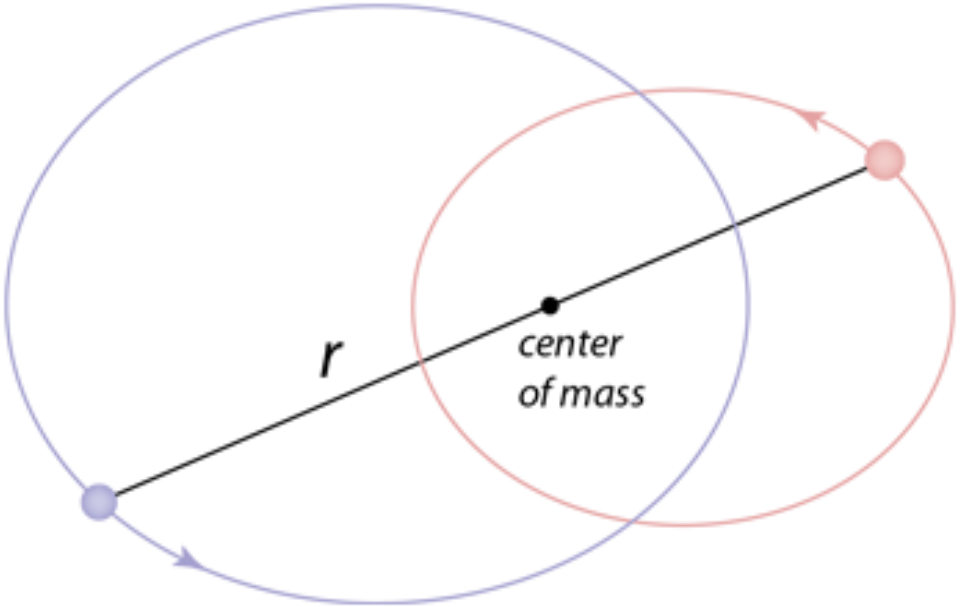
$$T^2 = \frac{4\pi^2 a^3}{GM_\odot},$$

CENTRAL FORCE: GRAVITY - REDUCED MASS



$$\mathcal{L} = \frac{1}{2} m_1 \dot{\mathbf{r}}_1^2 + \frac{1}{2} m_2 \dot{\mathbf{r}}_2^2 - V(|\mathbf{r}_1 - \mathbf{r}_2|)$$

CENTRAL FORCE: GRAVITY - REDUCED MASS



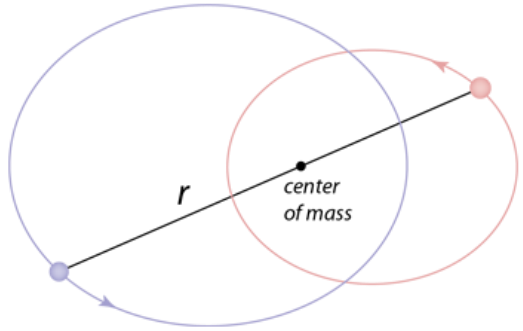
$$\mathcal{L} = \frac{1}{2} m_1 \dot{\mathbf{r}}_1^2 + \frac{1}{2} m_2 \dot{\mathbf{r}}_2^2 - V(|\mathbf{r}_1 - \mathbf{r}_2|)$$

$$\Downarrow \quad \frac{1}{\mu} \equiv \frac{1}{m} + \frac{1}{M_\odot}$$

$$\mathcal{L} = \frac{1}{2} \mu \dot{\mathbf{r}}^2 + \frac{\alpha}{r} \qquad (\text{where } \alpha \equiv G M_\odot m)$$

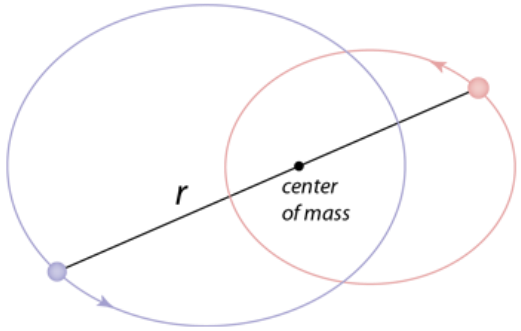
CENTRAL FORCE: KEPLER'S LAWS - REVISITED

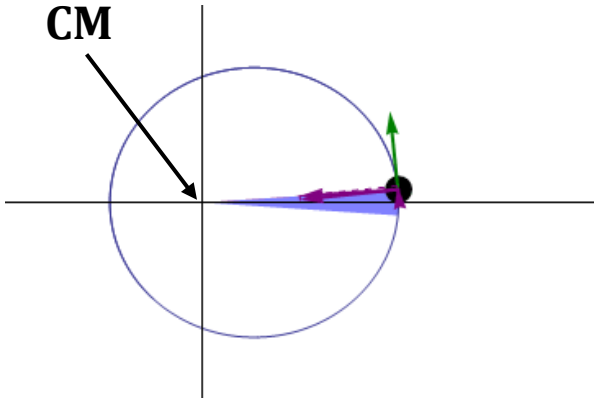
- **First law:** The elliptical statement in the first law is still true, but with the CM (not the sun) located at a focus. The sun also travels in an ellipse with the CM at a focus.



CENTRAL FORCE: KEPLER'S LAWS - REVISITED

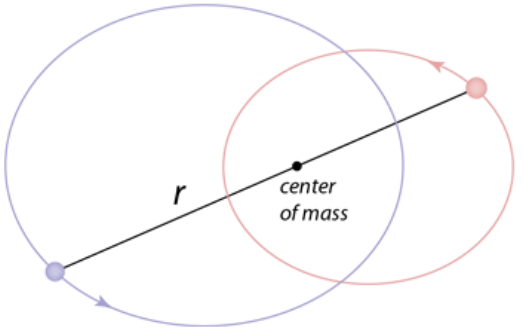
- **First law:** The elliptical statement in the first law is still true, but with the CM (not the sun) located at a focus. The sun also travels in an ellipse with the CM at a focus.
- **Second law**: In the second law, we need to consider the position vector from the CM (not the sun) to the earth. This vector sweeps out equal areas in equal times, because the angular momentum of the earth (and the sun, too) relative to the CM is constant.

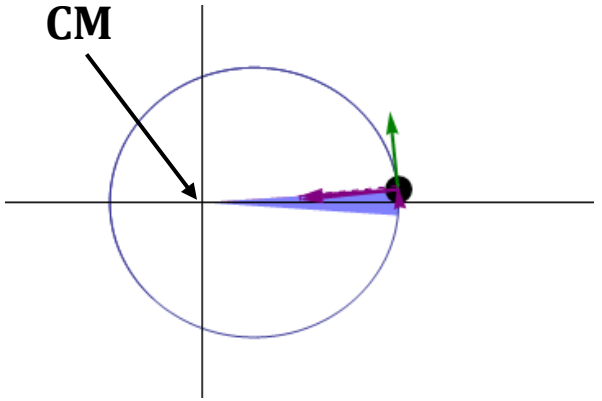




CENTRAL FORCE: KEPLER'S LAWS - REVISITED

- **First law:** The elliptical statement in the first law is still true, but with the CM (not the sun) located at a focus. The sun also travels in an ellipse with the CM at a focus.
- **Second law**: In the second law, we need to consider the position vector from the CM (not the sun) to the earth. This vector sweeps out equal areas in equal times, because the angular momentum of the earth (and the sun, too) relative to the CM is constant.
- **Third law**: The period of the earth's orbit (and the sun's, too) is the same as the period of the orbit of our hypothetical particle of mass μ orbiting around a fixed origin under the influence of the potential $-\alpha/r$

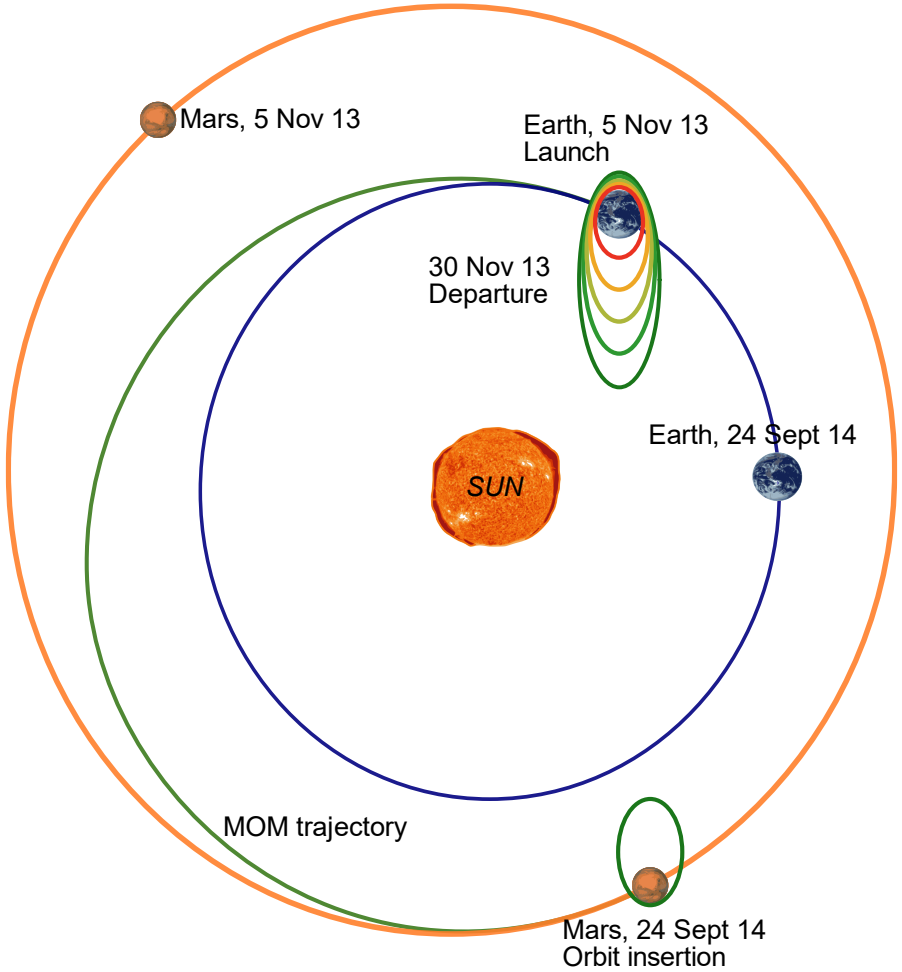


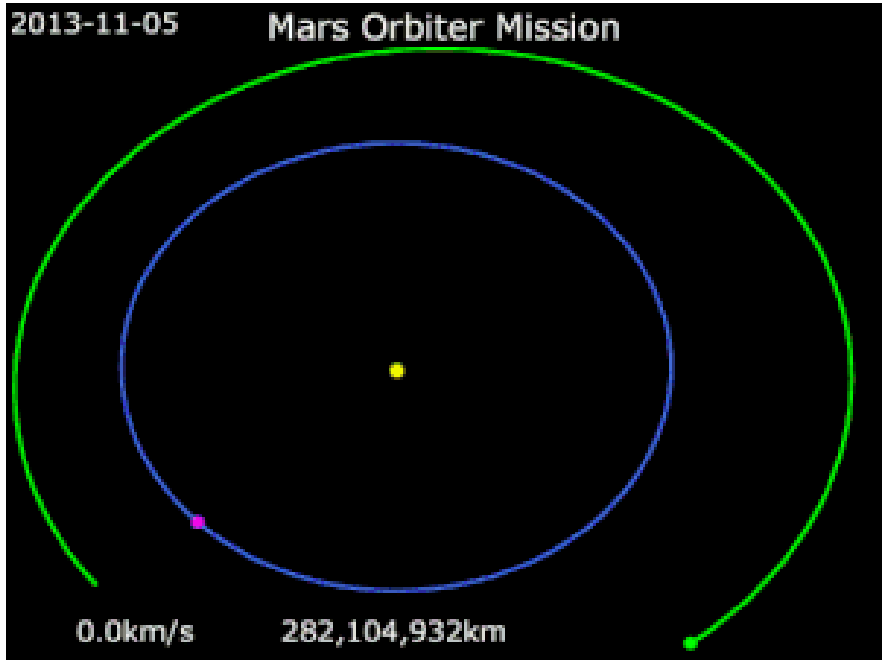


$$T^2 = \frac{4\pi^2 a_\mu^3 \mu}{GM_\odot m} = \frac{4\pi^2 a_\mu^3}{GM} \quad \Rightarrow \quad T^2 = \frac{4\pi^2 (M/M_\odot)^3 a_E^3}{GM} = \left(\frac{M^2}{M_\odot^2}\right)\frac{4\pi^2 a_E^3}{GM_\odot}$$

$$\mu \equiv M_\odot m/(M_\odot + m) \equiv M_\odot m/M \qquad a_E = (M_\odot/M)a_\mu$$

ORBITAL DYNAMICS: HOHMANN TRANSFER





TUTORIAL 8

5. A satellite's crew in a circular orbit of radius R_1 wishes to transfer to a circular orbit of radius $2R_1$. It is achieved using two successive boosts, as shown in the figure below. Firstly, the satellite is boosted at the point P into the elliptical transfer orbit 2, just large enough to reach out to the larger radius. Secondly, on reaching the required radius at P', (the apogee or farthest point of the transfer orbit), the satellite is boosted into the desired circular orbit 3.

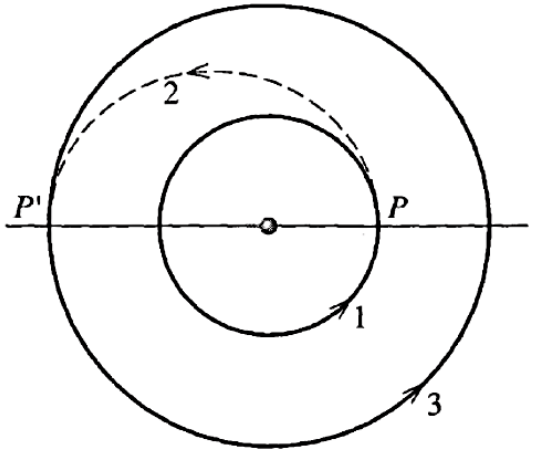
   (a) What is the factor by which must the speed of the satellite be increased in each of these two boosts?

   (b) What is the factor by which does the satellite's speed increases as a result of the complete maneuver?

$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon \cos\theta),$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}}$$

$$\alpha \equiv GM_{\odot}m$$



TUTORIAL 8

1. A particle of mass m, angular momentum L and energy E is moving under the influence of a force-field characterised by potential energy $U(r) = -C/r^n$, where n is a positive integer and $C > 0$.

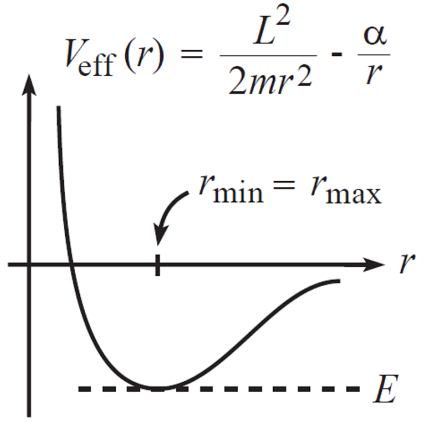
   (a) Sketch the effective potential $U_{\text{eff}}(r)$ as a function of r, for $n = 1, 2, 3$.

   (b) Show that a *stable* circular orbit is possible only for $n = 1$.

   (c) Let E_0 be the energy corresponding to the (stable) circular orbit for $n = 1$, and, $\Delta E = E - E_0$. For $\Delta E \ll |E_0|$, the radial coordinate should execute *small oscillations*. Determine the period for the same.

- **Circle** ($\epsilon = 0$)

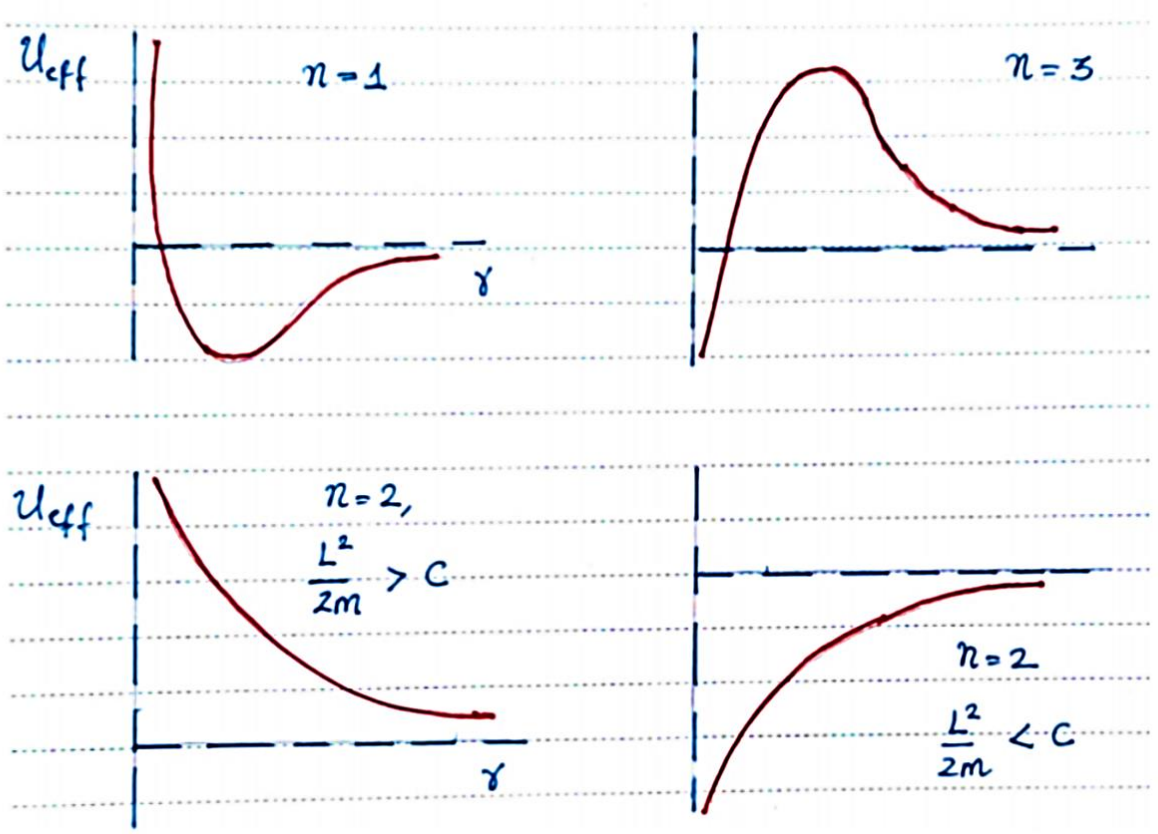
$$E = -m\alpha^2/2L^2$$



TUTORIAL 8

(a) Given $U(r) = -\frac{C}{r^n}$. Therefre the U_{eff} is

$$U_{eff} = -\frac{C}{r^n} + \frac{L^2}{2mr^2}$$



TUTORIAL 8

3. A particle of mass m is moving in an elliptic orbit (with eccentricity ϵ) in a central potential $U(r) = -k/r$, where k is a positive constant.

   (a) Show that the **Laplace-Runge-Lenz** vector

$$\vec{A} = \left(-\frac{1}{2mE}\right)^{1/2} [mk\,\hat{r} + \vec{L} \times \vec{p}]$$

   is a constant of motion. Can you provide a geometric interpretation for this vector? Hint: First evaluate it explicitly.

RECAP: SPHERICAL COORDINATE SYSTEM

Position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\Rightarrow \boxed{\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}}$$

$$\hat{r} = \frac{\partial\vec{r}/\partial r}{\left|\partial\vec{r}/\partial r\right|} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \frac{\partial\vec{r}/\partial\theta}{\left|\partial\vec{r}/\partial\theta\right|} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\phi} = \frac{\partial\vec{r}/\partial\phi}{\left|\partial\vec{r}/\partial\phi\right|} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} = \left(\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}\right)\dot{\theta} + \sin\theta\left(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y}\right)\dot{\phi}$$

$$= \dot{\theta}\hat{\theta} + \sin\theta\,\dot{\phi}\,\hat{\phi}$$

$$\Rightarrow \boxed{\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}}$$

RECAP: SPHERICAL COORDINATE SYSTEM

Position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\Rightarrow \boxed{\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}}$$

$$\hat{r} = \frac{\partial\vec{r}/\partial r}{\left|\partial\vec{r}/\partial r\right|} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \frac{\partial\vec{r}/\partial\theta}{\left|\partial\vec{r}/\partial\theta\right|} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

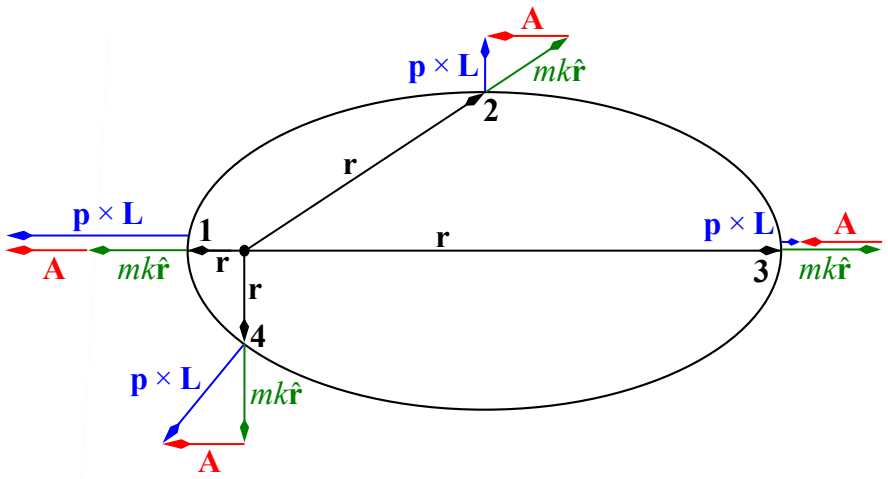
$$\hat{\phi} = \frac{\partial\vec{r}/\partial\phi}{\left|\partial\vec{r}/\partial\phi\right|} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

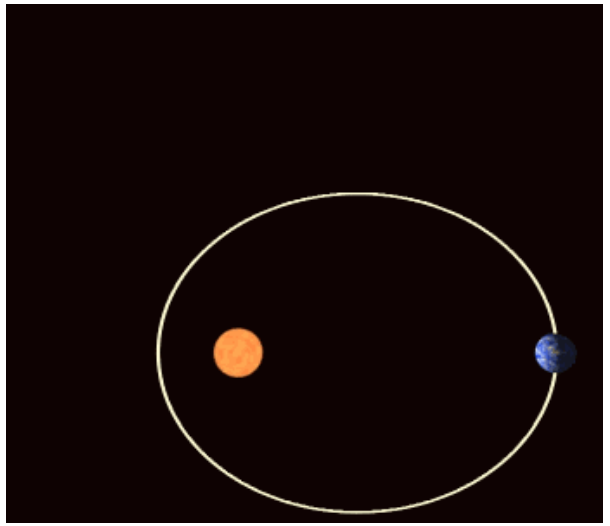
$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} = \left(\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}\right)\dot{\theta} + \sin\theta\left(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y}\right)\dot{\phi}$$

$$= \dot{\theta}\hat{\theta} + \sin\theta\,\dot{\phi}\,\hat{\phi}$$

$$\Rightarrow \boxed{\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}}$$





TUTORIAL 8

4. A particle of mass m moves in a potential given by $V(r) = r^k$. Let the angular momentum be L.

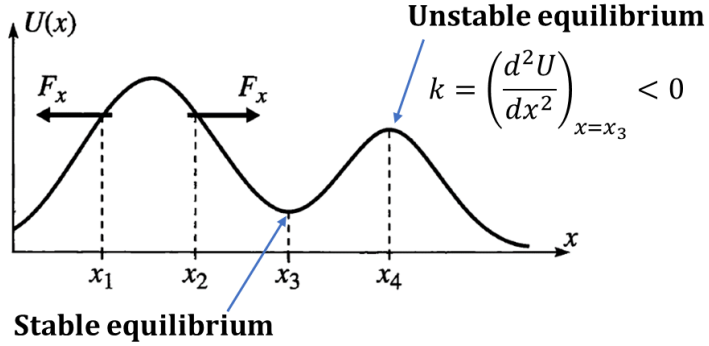
   (a) Find the radius, r_0, of the circular orbit.

   (b) If the particle is given a tiny kick so that the radius oscillates around r_0, find the frequency, ω_r, of these small oscillations in r.

   (c) What is the ratio of the frequency ω_r to the frequency of the (nearly) circular motion, $(\omega_\theta \equiv \dot{\theta})$? Give a few values of k for which the ratio is rational, that is, for which the path of the nearly circular motion closes back on itself.

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{\alpha}{r}$$

$$\mathbf{F}(\mathbf{r}) = -\nabla V(r) = -\frac{dV}{dr}\hat{\mathbf{r}}$$



$$k = \left(\frac{d^2U}{dx^2}\right)_{x=x_3} > 0 \qquad \omega = \sqrt{\frac{k}{m}}$$

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